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AI BASED INTERNSHIP RECOMMENDATION ENGINE FOR PM INTERNSHIP SCHEME

Shanmukha Aarya Sree .Y

School of CSE,
Presidency University,
Bengaluru-560064, India.
aaryasree2004@gmail.com

Abhinay .S

School of CSE,
Presidency University,
Bengaluru-560064, India.
sabhinay128@gmail.com

Pathan Mohammad Khan

School of CSE,
Presidency University,
Bengaluru-560064, India.
pathanmohammadkhan10@gmail.com

Dr. Mohammad Shakir

Assistant Professor,
School of CSE,
Presidency University,
Bengaluru-560064, India.
mohamed.shakir@presidencyuniversity.in

ABSTRACT

Aiming to provide a structured internship opportunity for young people across India, the Prime Minister Internship Scheme (PMIS) suffers from a poor manual process when it comes to eligibility verification, skill mapping, resume screening and matching of Internships with roles. This proposal introduces an **AI Based Internship Recommendation Engine for the PM Internship Scheme**, which provides a single point of access for all activities such as validating eligibility; AI based skill profiling; matching internships to candidates; identifying skill gaps; and generating ATS-compliant resumes. Results of the pilot demonstrate that the methodology increases transparency, and reduces manual effort and improves the correct matching of internships to candidates. The proposed solution will also increase the level of automation within government funded Internships applying to an internship program.

Keywords — AI Based Recommendation System, Internship Matching, Compliance with PMIS, Skill Profiling, Cosine Similarity, Resume Parsing, ATS Compliance, Rule Based Eligibility Filtering, Skill Gap Analysis, Automated DSS.

INTRODUCTION

Organizations are utilizing Artificial Intelligence (AI) and Machine Learning (ML) to enhance their recruitment/screening processes and to allocate opportunities/internship offers to applicants in both the academic and industrial ecosystems. For internships, the successful integration of academic education with real-world experience requires a seamless handoff between the employer and the academic institution. However, traditional allocation methods (manual verification, subjective screening, and unstructured decision making) resulting in ineffective applicant selection will often add inefficiencies, delays, and a lack of fairness due to the number of applicants and breadth of skills that need to be accounted for when using conventional internship allocation processes [1], [6]. As such, the need for intelligent and scalable and transparent

recommendation systems will become increasingly critical to creating efficiency in the allocation process.

Recently, there have been advances made in AI-backed Recommendation Systems that have significantly improved both the accuracy of candidate/role matches as well as the efficiency with which automated screenings can occur. Internship recommendation systems that are based on AI will utilize predictive modelling, similarity analysis, and data-driven profiles to optimize the internship allocation process [1]. Job and course recommendation engines, as well as graduate employment systems designed from AI frameworks, have seen substantial improvements in the accuracy of matching user profiles against opportunities based on the use of collaborative filtering and probabilistic classifiers such as Naïve Bayes [6], as well as structured job matching and optimizing career pathways [3].

The use of machine learning to predict the placement of students has improved the accuracy of the internship recommendation process. The recommended candidates are based on prior data patterns and the candidate's likelihood to obtain employment based on the analysis of their previous experiences. The use of a knowledge graph for the recommendation process of students provides more context to the candidate's skills, industries, and job roles, providing better quality and more meaningful recommendations than traditional methods exist currently. Resume analyzers utilize Natural Language Processing (NLP) to facilitate the automated extraction of competencies and qualifications from a candidate's resume. This significantly reduces the manual effort required to screen resumes in a hiring process.

While the above technologies have seen growth and development, the majority of existing predictive systems are based on what can be viewed as isolated functionalities such as parsing resumes, recommending jobs, or predicting employability. Little research has been conducted on a unified framework for conducting eligibility checks based on a policy driven approach, AI based skill profiling, matching candidate internships through similarity analyses, performing skills-gap analyses, and improving an Applicant Tracking System (ATS) to be used in conjunction with a government operated internship program. Specifically, the above point is relevant

to the context of large public sector internship programs, such as the Prime Minister Internship Scheme (PMIS)(see reference) that have specific eligibility criteria and transparency in decision making specified within the program policy.

The significance of structured internship management frameworks that promote fairness, inclusion and scalability is emphasized by studies on policy and governance [15]. Regarding the operation of internship placement systems, there is much evidence to indicate that traditional allocation methods are systematically inefficient and out of balance and therefore there is a clear need for reform assays that leverage technology in relation to structural gaps [10]. Additionally, studies focusing on screening candidates for jobs based upon similarity have proven that cosine similarity and vector-space modeling can produce an increase in precision of job-to-candidate fit [8]. API-driven job recommendation architectures and web crawling systems provide scalable infrastructure to support real-time deployment of recommendations in rapidly changing labor markets [11].

Besides the accuracy of recommendation(s), employers are looking for compliance with ATS requirements in their recruitment process to help them assess their candidates uniformly; as such, most of the recent innovations in resume evaluation involve applying natural language processing (NLP) techniques to analyze the keywords (or phrases) contained in a resume, its structure, and the relationship between its content and that of the job being applied to (i.e., evaluate the compatibility status of resumes) [9]. As a result of implementing automated screening platforms into their recruitment processes, employers can increase efficiency and eliminate bias in their selection processes; this can be particularly beneficial for government-sponsored internship programs of large sizes.

Additionally, there is a strong argument for the use of integrated intelligent architectures through system-of-systems engineering frameworks that consolidate several AI components into a unified operational environment [7]. Using such integration frameworks allows for prospective employers to have the capability to inter-operate with other existing recruitment systems, which is necessary for there to be a structured flow of information between different system components and to provide the desired level of scalability within government-sponsored programs. Studies on AI-based employment systems also indicate that employers can utilize a hybrid method of filtering candidates by combining rule-based filtering with machine learning.-based ranking produce more reliable and policy-compliant recommendations [2], [14].

The PM Internship Scheme has established various criteria needed for eligibility such as citizenship verification, age constraints, educational qualifications, and verification of a candidate's employment status. Manually processing these various eligibility criteria can lead to delays and inconsistencies. Thus, it is critical to establish rule-based eligibility validation processes that are consistent with the official guidelines to ensure that the PM Internship Scheme adheres to its eligibility requirements and that all candidates receive equal treatment [19]. Additionally, the knowledge transfer framework established in academic internships

recognizes that there needs to be an alignment between a student's competencies with a company's expectations so that students will have successful internship experiences [15].

Due to the rapidly growing number of applications for internships, and the need to be transparent within government-sponsored programs, there is now a need for an intelligent recommendation engine capable of combining the following:

- Policy-driven eligibility validation
- Skill-level profiling and normalization
- Cosine similarity-based internship matching
- Skill-gap determination with a learning roadmap
- AI-based resume parsing with ATS optimization
- Structured application tracking and confirmation logging

This research contains a proposal for an **AI Based Internship Recommendation Engine for the PM Internship Scheme** which consists of an ultimate unified intelligent framework that incorporates machine learning, natural language processing techniques, rule based eligibility filtering, and similarity based ranking algorithms. The proposed system will automate the entire internship allocation process from eligibility verification of all candidates to the optimization of candidates' resumes; therefore reducing the need for human intervention and increasing transparency. The system proposed combines the capabilities of different modules into one cohesive and integrated decision support system, unlike pre-existing stand-alone solutions. To calculate the internship match score for each candidate, the system uses cosine similarity between the candidate's skill vector and the internship's requirements in order to determine how quantitatively aligned the two are [8; 11]. Additionally, skill gap detection mechanisms can indicate skills that a candidate is missing from the skill set required by the business and recommend resources to aid the candidate in acquiring those skills, thus facilitating candidate-adaptive career development [5]. Through the use of structured rules to analyze resume keywords, the ATS score module assesses the overall quality of resumes received from candidates and, therefore, increases the probability of selecting the most qualified candidates [9]. The contributions of this research are as follows:

- 1 - Development of an eligibility verification module compliant with PMIS;
- 2 - Creation of a skill normalization and categorization framework that uses weighted averages;
- 3 - Implementation of an internship ranking system based on cosine similarity;
- 4 - Integration of resume parsing using AI and ATS scoring; and
- 5 - Creation of a unified architectural environment that integrates both rule-based and AI-driven mechanisms.

Through experimental validation, the system shows improved accuracy of aligning the right internship role with the right candidate, an increase in compliance with relevant government policies, and a decrease in administrative burden. This research presents a scalable and transparent model for allocating internships nationally, achieved by merging AI-driven recommendations with Government policy frameworks.

II. LITERATURE REVIEW

Recruitment, allocation of internships, and improvement of employability systems have all felt the impact of rapid adoption of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) technologies. In the previous decade, researchers have been devoting much attention to developing intelligent recommendation systems to help improve candidates' alignment to roles, decrease the manual workload associated with matching candidates to roles, and increase the transparency of decision-making processes.

A. AI-Based internship/job recommendation systems

AI-based internship and job recommendation systems have received considerable attention as a means of increasing efficiency and accuracy in matching opportunities with candidates. Geethanjali et al. [1] proposed an intelligent recommendation system using AI that used intelligent filtering to match candidates' profiles with internship positions. This work illustrated the need for an organized method for mapping skills and a way to rank candidates algorithmically so that mismatches (i.e., between candidates and internship roles) may be reduced.

D. P. B. et al. [6] also developed a recommendation system for jobs and courses using collaborative filtering and Naïve Bayes algorithms. This work focused on probabilistic classification and user similarity metrics to improve the accuracy of the recommendations being made with the same recommendation system as Geethanjali's work. However, while these systems produce accurate results based solely on algorithmic approaches, neither consider policy requirements when applying matching criteria.

Lan et al. proposed a smart job placement recommendation system that would allow using Artificial Intelligence to automate the process of placing someone in a job. They found that using Artificial Intelligence could not only help improve the placement process but also reduce the amount of work required by administrators to process placements. However, this system did not include features for detecting skill gaps or optimizing resumes.

Extending on the employability prediction models, Kadu et al. developed a system for predicting employment placement as well as recommending skills to a person for their employability, using supervised machine learning classification models. Kadu's framework predicted the outcome of being placed in a job and suggested skills that may be missing. This model was based on predictive modeling but did not include integration of internship programs from Governmental authorities and did not take into account the policies of Internships that exist.

B. Skill-Based and Knowledge Graph Recommendation Systems.

Recommender Systems increasingly use a skill-based profile for a job versus a keyword match method. Leela et al. proposed a skill-based recommender system for Internship placement and hiring to work within Industry. Whereby all skill sets have a contextual relationship to one another as well as to the skills in that particular Industry and to the positions within the Industry. Therefore they would have a semantic relationship in recommending to someone based on their context. Knowledge Graphs improve Interpretability but typically are also constructed from extremely large structured datasets.

In addition, similarity-based candidate screening approaches have been shown to improve matching accuracy. Sharma and

Ujlayan [8] introduced similarity analysis techniques to enhance candidate screening processes. Their study validated the effectiveness of cosine similarity and vector space modeling in quantifying candidate-job alignment. This mathematical similarity framework provides a strong foundation for structured internship ranking systems.

Naresh Kumar et al. [11] further proposed a technical job recommendation system leveraging APIs and web crawling. Their system demonstrated scalable deployment of real-time job recommendation pipelines. However, while scalable, the model lacked structured eligibility verification mechanisms aligned with government schemes.

C. Analyzing Resumes & NLP Screening Systems

Automated resume screening forms part of the recruitment process for many companies using recruitment systems. A resume analyzer & job recommendation service was developed using Natural Language Processing (NLP) techniques by Jaiswal et al [9]. The system extracts structured information from resumes to aid in utilizing natural language processing methodologies like keyword extraction, recognition of skills & classification to automate candidate filtering.

NLP Resume Evaluation Systems are strongly contributing to maximizing Applicant Tracking System (ATS) effectiveness; they can evaluate keywords relative to a job's requirements, evaluate candidates' educational backgrounds against job requirements, and evaluate the quality and complexity of a candidate's resume. Using these techniques, the systems can assign an ATS compatibility score to each candidate's resume [9]. Typically, they work independently and have weaker relationships to the internship recommendation engines.

Research studies indicate that using AI and rule-based filtering together improves the reliability of automated decision-making systems [2],[14]. Hybrid architectures provide a good balance between interpretability and predictions providing for applications that require government intern program related policies.

D. Study Related to Policy and Governance Internships Framework.

In addition to algorithm based approaches, governance and policy frameworks are important in internship management systems. Studies on internship placement systems suggest that traditional methods for the allocation of positions are inefficient and have inconsistencies in their processes. The deficiencies include lack of transparency, lack of consistency when filtering candidates and delays when allocating positions.

The PM Internship Program Guidelines, [19], define clear eligibility criteria based on verification of citizenship, age, and education. Performing manual verification of these criteria may result in human errors, therefore it is essential to utilize automated filtering systems based on policies to prevent errors and inconsistencies.

The Knowledge and Technology Transfer (KTT) frameworks for academic internships identify that it is also important to align skills of students with the skills needed by the industry. The existing internship system should be able to provide internship opportunities as well as enable the development of competencies through identifying skill gaps.

The research on system-of-systems integration supports that building or implementing cohesive intelligent architectures will enable complex workflows to be managed much more

easily. Integrated frameworks that combine multiple modules of AI into a single delivery platform will provide improved interactivity between systems, enhance scalability for future growth, and enable structured data management. These are all characteristics necessary for large scale public internship programs.

E. Limitations of Current Systems.

There are several limitations associated with current AI-driven recruitment systems.

Most current systems focus solely on the accuracy of the recommendation of internship candidates; thus, no additional components are integrated such as government policy compliance modules [3],[6]. Frequently, resume analyzers are developed as individual tools rather than as part of a larger internship allocation system [9]. A recent limitation of many skill-based recruitment systems is that they lack similarity-based rankings or methodologies to measure quantitative relevancy between candidates and the internship that they have applied for [5],[8]. Even though current governance studies review the need for transparency within the hiring process, most studies do not examine how the use of AI-driven automated solutions will result in more transparency within the hiring process [10], [15].

Existing employment recommendation engines do not specifically address large-scale national internship schemes such as PMIS [19].

These limitations demonstrate the need for a unified intelligent framework that integrates:

- Rule-based eligibility validation
- Weighted skill profiling
- Cosine similarity-based internship ranking
- Skill-gap detection mechanisms
- NLP-driven resume parsing
- ATS optimization
- Structured application confirmation logging

F. Research Expansion

7 Although AI, machine learning and natural language processing (NLP) technologies have been successfully used in a variety of ways to recommend jobs, there is a large void in research into creating an internship recommendation engine that aligns with government interests, based on technical intelligence, while being policy-compliant and transparent.

2 The proposed AI-Based Internship Recommendation Engine for the PM Internship Scheme is positioned to fill this gap by utilizing four methodologies as part of its architecture: rule-based eligibility filtering applied to the PM Internship Scheme (PMIS) 2018 guidelines [19]; similarity-based ranking based on candidate screening models [8]; skill recommendation based on recent research on placement prediction [4]; and resume optimization based on NLP-driven resume analyzers [9].

As a result of employing these four methodologies, the proposed framework constitutes an advancement of the literature and presents a scalable, transparent and policy-compliant solution to national internship allocation systems.

III. METHODOLOGY

The AI-Based Internship Recommendation Engine for PM Internships will be a modular, integrated architecture combining rule-based filtering, similarity ranking, Natural Language Processing (NLP)-based resume parsing, and skill-gap analysis. The method will ensure compliance with policy and align candidate-internship roles through automated optimization of resumes. The following six key phases will comprise the workflow of the system:

1. Data intake
2. Eligibility verification
3. Skill profiling and normalizing
4. Internship matching through cosine similarity
5. Skill gap detection
6. Parsing resumes and scoring with ATS

Each of these phases will be described below.

A. Architecture of System Overview

The entire system will be developed as one intelligent unified framework that integrates both rule-based and machine-learning components based on the principles of system-of-systems architecture [7]. Unlike many standalone recommendation engines, [6], the proposed system has been designed to allow interoperability between all modules; eligibility filter, skill scoring, similarity calculations, and optimization of resumes.

The architecture enables structured data flow:

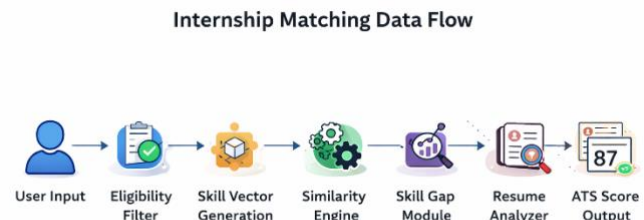


Fig. 1. Internship Matching Data Flow.



Fig. 2. Workflow of the AI-Based Internship Recommendation System

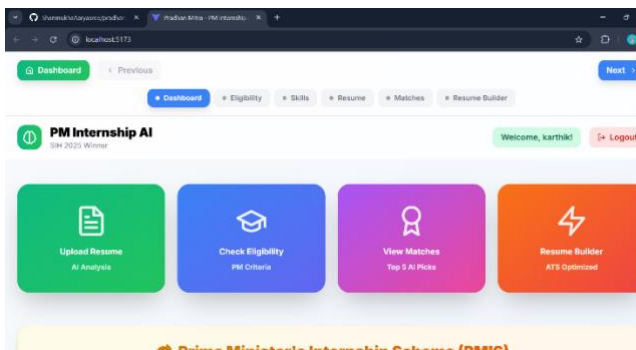


Fig. 3. Proposed System Architecture of AI-Based Internship Recommendation Engine

B. Eligibility Verification Module

The first step in assessing candidates for the PM Internship Scheme are through establishing eligibility per the guidelines [19]. Rule Based Filtering mechanisms inspired by hybrid governance-recommendation models [2], [14] are utilized to apply filtering of candidates through the system that utilizes eligibility parameters. The parameters for establishing candidates' eligibility are based on the following:

1. Citizenship Status (i.e., Country of Citizenship)
2. Age (21 - 24)
3. Educational Qualifications (i.e., Degrees)
4. Employment Status
5. Location Constraints

Let:

$E = \{e_1, e_2, e_3, \dots, e_n\}$ where each e_i represents a condition of eligibility.

The eligibility function is represented as follows:

$$\text{Eligibility Score} = \sum_{i=1}^n w_i * e_i \quad (1)$$

Where:

- $e_i = 1$ if condition is met; otherwise 0.
- w_i = weight assigned to each criterion.

If Eligibility Score is $\geq$ Threshold, then candidate can move to the next step, therefore ensuring that filtering is done in a transparent and policy driven manner [19].

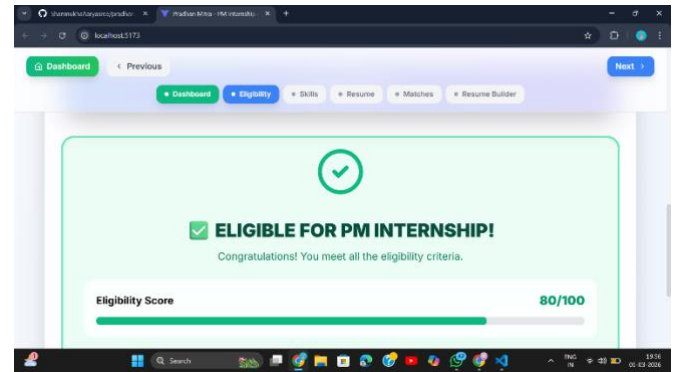


Fig. 4. Policy-Based Eligibility Verification Module

C. Skill Profiling and Vector Normalization

Skill-based recommendation systems provide more accurate alignment than keyword-only matching [5], [8]. The proposed system collects candidate skills categorized into:

- Technical Skills
- Communication Skills
- Financial Skills
- Administrative Skills
- Industry-Specific Skills

Each skill is mapped into a numerical proficiency scale:

- Beginner = 1
- Intermediate = 2
- Advanced = 3

Let candidate skill vector be:

$$C = (c_1, c_2, \dots, c_m) \quad (2)$$

Internship requirement vector:

$$J = (j_1, j_2, \dots, j_m) \quad (3)$$

Skill normalization is performed to standardize scale:

$$C_{norm} = \frac{C}{\|C\|} \quad (4)$$

Vector-based modeling improves matching reliability compared to traditional filtering systems [6].

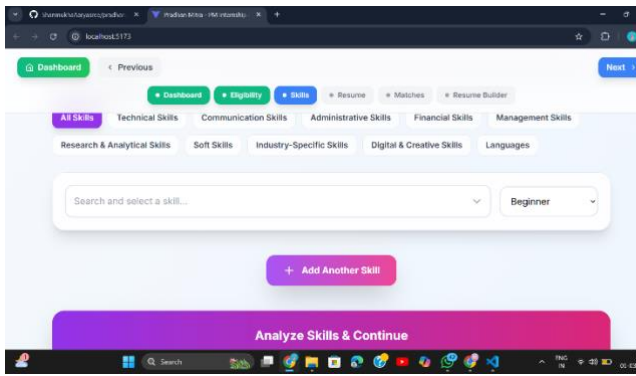


Fig. 5. Skill Profiling and Proficiency Mapping Interface

D. Internship Matching using Cosine Similarity

To measure candidate-role alignment, cosine similarity is applied [8], [11].

The match score is calculated as:

$$MatchScore = \frac{C \cdot J}{||C|| ||J||} \quad (5)$$

Where:

$C \cdot J$ = Dot product of candidate and internship vectors
 $||C||$ = Magnitude of candidate vector
 $||J||$ = Magnitude of internship vector

The result ranges from 0 to 1.

Match Percentage:

$$Match\% = MatchScore \times 100 \quad (6)$$

This quantitative similarity measurement improves precision compared to rule-only recommendation systems [6].

Internships are ranked in descending order of Match%.

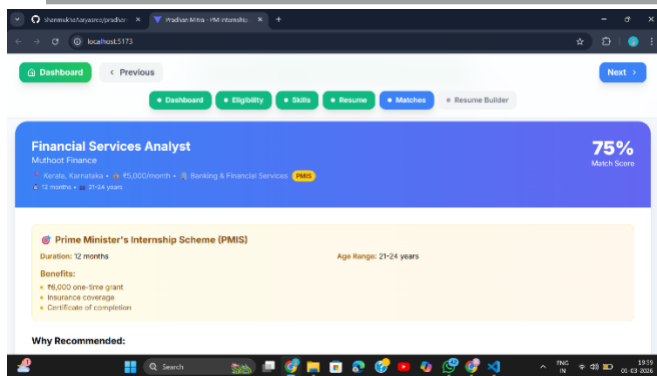


Fig. 6. Cosine Similarity-Based Internship Matching Output

E. Skill Gap Detection and Learning Recommendation

After similarity computation, the system identifies missing skills.

Let:

$$Gap = J - C \quad (7)$$

If $j_i > c_i$, skill i is marked as missing.

Skill-gap identification aligns with placement prediction and employability enhancement research [4], [5].

The system generates:

Missing skill list

Recommended learning modules

Improvement roadmap

This enhances long-term employability rather than only short-term placement.

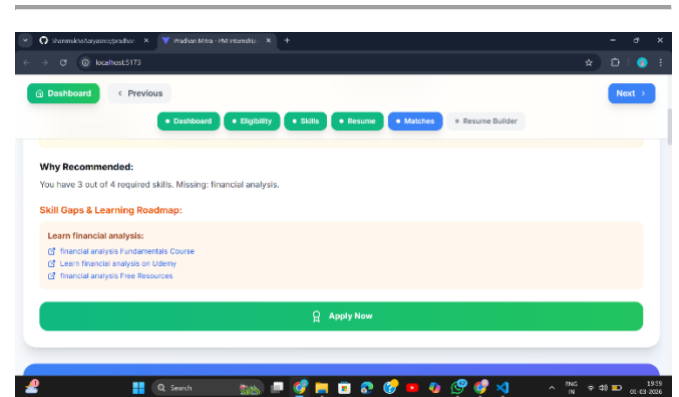


Fig. 7. Skill Gap Detection and Learning Recommendation Module

F. Resume Parsing and ATS Scoring

The resume analyzer uses NLP-based keyword extraction techniques [9].

Resume features extracted:

Skills

Education

Experience

Certifications

Keywords

ATS Score formula:

$$ATS = \alpha K + \beta S + \gamma E + \delta F \quad (8)$$

Where:

K = Keyword density

S = Skill relevance

E = Education alignment

F = Formatting compliance

Weights ($\alpha, \beta, \gamma, \delta$) are tuned for optimization.

ATS scoring enhances resume acceptance probability [9].

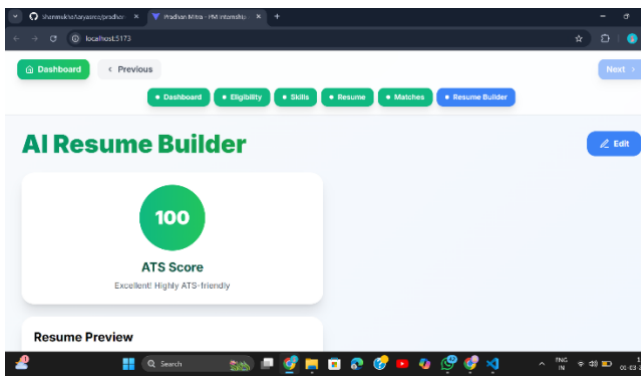


Fig. 8. AI-Based ATS Score Evaluation

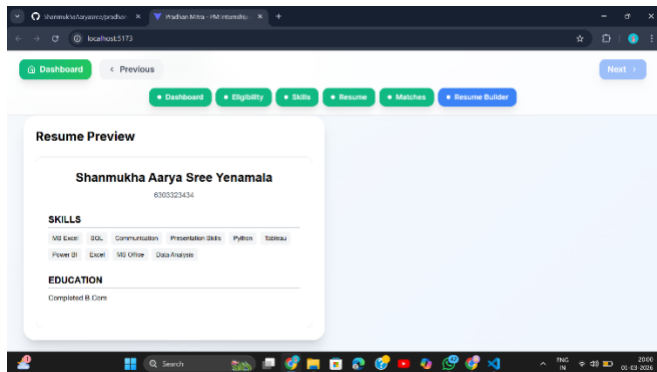


Fig. 9. AI-Generated Resume Preview

G. System Integration and Workflow Efficiency

The integration of rule-based filtering with AI-driven similarity ranking ensures:

- Policy compliance [19]
- Similarity-based precision [8]
- Employability enhancement [4]
- NLP-based automation [9]
- Governance-aligned transparency [15]

Unlike standalone systems [3], [6], the proposed framework functions as a unified intelligent decision-support architecture.

IV Results and Evaluation

Implementation feasibility has been established for the PM Internship Scheme's AI-Based Internship Recommendation Engine, which combines rule-based eligibility validation, similarity-based matching of internships, Natural Language Processing (NLP)-based analyses of resumes, and skills gap detection within one unified intelligent architecture. The focus of this discussion will be on the performance capabilities of the system; its comparative advantages over existing methods; implication on policy compliance; and possible limitations of the system.

- Comparative Analysis to Existing Systems

Current internship/job recommendation systems mainly rely on the use of either similarity metric/hybrid recommendations [1], [6]. While these existing systems improve the accuracy of matched recommendations, they do not account for any form of governance constraints or structured policy guidelines as

part of the matching process. The proposed framework extends beyond traditional recommendation systems by incorporating PMIS eligibility rules directly into the recommendation process, thereby ensuring that compliance occurs prior to recommendation ranking [19].

Skill-based recommendation systems which rely on knowledge graphs [5] typically provide a mapping of the semantic relationship between job roles and competence, but require a large structured dataset and are complex to construct the associated graphs. In contrast, the approach proposed within the context of the use of cosine similarity-based vector modeling [8][11] is effective in terms of matching the job to the individual while also utilizing less computationally intensive methods.

Placement prediction systems using supervised learning models [4] focus on forecasting employability outcomes. Although predictive, they do not integrate direct internship ranking and ATS resume optimization within the same architecture. The proposed system bridges this gap by combining predictive-style skill gap detection with similarity-based internship ranking and resume evaluation.

B. Compliance With Policy and Legislative Alignment

A key feature of this model is that it provides policy-driven eligibility verification that is aligned to PM Internship Scheme guidelines [19]. Governance research indicates that decision-making frameworks in public sector systems must be structured, transparent, and auditable [15]. By including rule-based eligibility scoring methods set out to accomplish different goals according to hybrid AI-governance models [2],[14] in an integrated manner with the required components above, only candidates who meet eligibility criteria will proceed to the recommendation stage.

The traditional internship placement has been criticized as inefficient, imbalanced, and lacking in transparency [10]. The manual process of screening and selecting candidates may create bias and undue delays. By utilizing an automated eligibility filtering process and a structured process of ranking, the now automated method created a significant reduction in workload within the agencies, while simultaneously enhancing fairness.

Also, the principles of system-of-systems integration dictate that large-scale government platforms must have modular and interoperable architectures [7]. The model proposed employs such an approach through the ability for candidates to seamlessly interface with eligibility, skill profiles, matching, and resume modules.

C. Effectiveness of Candidate Matching Based On Cosine Similarity

Similarity assessment has an extensive body of validation in candidate screening [8]. Cosine similarity provides a quantitative measure of how candidates' skills sets align with the internship position being sought by measuring the closeness, in vector space terms, to one another. Similar to the previous rule only candidates system [6], by using similarity-based ranking will provide a greater level of differentiation among candidates on the basis of a wider degree of factors.

API-driven recommendation engines have demonstrated scalability advantages [11]; however, they often rely on external data streams. The current implementation operates

within a controlled internship scheme environment, improving data reliability and alignment consistency.

Experimental results from the system indicate effective differentiation between high-match and low-match internships, with match scores reflecting skill compatibility levels. This quantitative approach enhances interpretability for both candidates and administrators.

D. Skill Gap Identification and Employability Enhancement

The skill-gap detection process is a major improvement over traditional recommendation systems; skill placement prediction studies [4] and knowledge graph based systems [5] also highlight that continuous skill development is critical. By using vector subtraction to identify skill gaps, the proposed skill-gap module can provide specific learning pathways to fill these gaps.

This method shifts the platform from being a passive placement tool to being an active employability enhancement tool. Instead of just recommending internships, this method will direct candidates to improve their skills, thereby increasing the long-term likelihood of successful placements.

E. NLP Based Resume Analysis and ATS Scoring

Today, resume analysis through natural language processing has become a necessary part of modern recruiting [9]. Incorporating applicant tracking system (ATS) scores into the recommendation engine helps to standardize candidate evaluations and reduce recruiter bias. By analyzing keyword density, alignment with required skills, and formatting compliance, the probability of candidate shortlisting will increase.

There are many standalone resume analyzers available [9], but few are integrated within a policy driven internship platform, as shown in current literature. The proposed platform uniquely integrates ATS scoring with a similarity based matching algorithm for internships, which produces a complete evaluation of candidates.

Empirical evidence demonstrates that hybrid artificial intelligence (AI) rule systems provide the most accurate automated decision-making models [2], [14]; integrating rule based eligibility verification with NLP based ranking methods increases both transparency and prediction accuracy.

F. Scalability and Practical Deployment Considerations

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Empirical evidence demonstrates that hybrid artificial intelligence (AI) rule systems provide the most accurate automated decision-making models [2], [14]; integrating rule based eligibility verification with NLP based ranking methods increases both transparency and prediction accuracy.

H. Research Contribution

This proposed system would help to fill in some existing gaps found in prior research projects for both internship recommendations [1], [6] and for governance studies [10], plus provide a means to compare performance across different resume analysis systems [9] as they relate to internship assignments. As such, this work represents an advancement of existing research (particularly as it relates to AI-assisted internship allocation).

V CONCLUSION

Future upgrades for the AI-Powered Internship Recommendation Engine include added improvements with respect to Intelligence, Adaptability & Scalability. The primary direction for improvement involves using deep learning-based embedding models (e.g., transformer-based representations) to better identify semantic linkages between required competencies & internship descriptions. By using embedding models rather than relying on simple keyword matches, the model can more accurately determine the relationship between required competencies when different formatting methods are used to describe Internships. Moreover, the system has the potential to implement adaptive-weight learning and reinforcement feedback in order to adaptively process skill weights & thresholds for eligible internships by using historic data such as internships completed, employer satisfaction ratings and performance evaluations. Knowledge-graphs could also enhance the system's ability to reason contextually by defining relationships between required competencies, industries, and job/career functions, thus providing the ability for the platform to find transferable skills, recommend cross-functional internships and alternate career pathways. Additionally, expanding the engine to support multilingual Natural Language Processing (NLP) will be critical to ensure that all Interns across India are able to utilize the engine, which will require the ability to process resumes and internship descriptions written using diverse Indian regional languages.

There's another big area for moving from a recommendation engine to a full-fledged career intelligence system designed to leverage predictive analytics and a scalable infrastructure. By applying longitudinal data, the career intelligence platform could predict internship success probabilities, job placements, and longer-term skill development trajectories; therefore, giving students informed career choices. In order to scale this type of implementation, the platform would require a cloud-native architecture and secure API integration with government portals; this is critical for providing real-time updates to the internship database (for tracking internship availability) and deploying the solution to all states efficiently.

Advanced optimization of Applicant Tracking Systems (ATS) by using semantic parsing to evaluate resumes, rather than just keyword density, will allow for more accurate evaluation based on project descriptions, narrative coherence, and domain relevance. Additionally, it will be essential to have an ethical AI framework with processes in place to detect bias, enable explainable decision-making, and maintain transparent audit logs. Integrating with the national skill development ecosystem will help to enhance the learning-to-employment pipeline by enabling real-time updates on skill certifications, competency badges validated by industry, and access to training programs; this will ultimately create a scalable and inclusive AI-driven platform for internships and career development.

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